When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer?

A multi-seed calibration boundary study: 6 transfer pairs × 2 architectures × 5 seeds

1 Experiment grid

Chapman–Shaoxing

CPSC2018

PTB-XL

6 directed pairs
patient-level splits
InceptionTime | ResNet1D
seeds 42–46

60 experiments

2 Deployment shift

(L2, eval-time, no retraining)

Sampling rate
500 → 250 → 125 Hz

Leads
12 → 6 → 3 → 2 → 1

Noise (BWL/MA/EM)
SNR 24 → −6 dB

Gain
×0.5 / ×2

3 Post-hoc recalibration

Raw model (no recalibration)

vs

Temperature Scaling (TS)

+ 7 further methods:
Platt · Isotonic · Vector · Matrix
Dirichlet · Saerens EM · BBSE

endpoint: $\Delta ECE_{OOD} = ECE_{raw} - ECE_{TS}$

patient-level cluster paired bootstrap
(BCa, B = 10,000)

4 When does TS pay off?

✓ OOD calibration benefit in 50/60 seed experiments (83.3%)

✓ OOD benefit ≈ 1.9 × ID benefit (+0.0154 vs +0.0082 ECE)

✗ Benefit is NOT predictable across architectures
(LOO R^2 CI ≤ 0.104 → registered failure branch)

✓ Discrimination-aware gate (C5): 7/12 strata pass both layers
(stratum mean; 24/60 cells, 1/12 strata pass in all 5 seeds)

Pre-specified protocol with registered, dated amendments; publicly archived with per-file checksums
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